

21.1

2d Geometry (1)

21.1.1 2d Geometry: UGC NET CSE | July 2018 | Part 2 | Question: 10



Consider the matrix $M = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ representing a set of planar (2D) geometric transformations in homogeneous coordinates. Which of the following statements about the matrix M is **True**?

- A. M represents first, a scaling of vector (2, 1) followed by translation of vector (1,1)
- B. M represents first, a translation of vector (1, 1) followed by scaling of vector (2,1)
- C. M represents first, a scaling of vector (3, 1) followed by shearing of parameters (-1,1)
- D. M represents first, a shearing of parameters (-1, 1) followed by scaling of vector (3,1)

ugcnetcse-july2018-paper2 computer-graphics 2d-geometry

Answer key

21.2

2d Transformation (1)

21.2.1 2d Transformation: UGC NET CSE | June 2019 | Part 2 | Question: 28



Consider the following statements regarding 2D transforms in computer graphics:

$S1 : \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is a 2×2 matrix that reflects (mirrors) only 2D point about the X-axis.

$S2 : A 2 \times 2$ matrix which mirrors any 2D point about the X-axis, is a rotation matrix.

What can you say about the statements $S1$ and $S2$?

- A. Both $S1$ and $S2$ are true
- B. Only $S1$ is true
- C. Only $S2$ is true
- D. Both $S1$ and $S2$ are false

ugcnetcse-june2019-paper2 2d-transformation

Answer key

21.3

3d Graphics (2)

21.3.1 3d Graphics: UGC NET CSE | July 2018 | Part 2 | Question: 9



Which of the following statements is/are **True** regarding the solution of the visibility problem in 3D graphics?

$S1$: The painter's algorithm sorts polygons by depth and then paints (scan - converts) each Polygon on to the screen starting with the most nearest polygon

$S2$: Backface Culling refers to eliminating geometry with backfacing normals

- A. $S1$ only
- B. $S2$ only
- C. Both $S1$ and $S2$
- D. Neither $S1$ nor $S2$

ugcnetcse-july2018-paper2 computer-graphics 3d-graphics

Answer key

21.3.2 3d Graphics: UGC NET CSE | June 2019 | Part 2 | Question: 29



In the context of 3D computer graphics, which of the following statements is/are true?

P : Orthographic transformations keep parallel lines parallel.

Q : Orthographic transformations are affine transformations.

Select the correct answer from the options given below:

- A. Both P and Q
- B. Neither P nor Q
- C. Only P
- D. Only Q

Answer key **21.4 3d Rotation (1)****21.4.1 3d Rotation: UGC NET CSE | September 2013 | Part 3 | Question: 64**

Consider the following transformation matrix for rotation(clockwise):

$$[T] = \begin{bmatrix} \cos \theta & \sin \theta & 0 & 0 \\ -\sin \theta & \cos \theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

This matrix rotates an object by an angle θ about:

- A. X-axis B. Y-axis C. Z-axis D. All of the above

ugcnetcse-sep2013-paper3 computer-graphics 3d-rotation

Answer key **21.5 Animation (1)****21.5.1 Animation: UGC NET CSE | December 2014 | Part 3 | Question: 13**

Which of the following categories of languages do not refer to animation languages ?

- A. Graphical languages B. General-purpose languages
C. Linear-list notations D. None of the above

ugcnetcse-dec2014-paper3 computer-graphics animation non-gatecse

Answer key **21.6 Anti Aliasing (1)****21.6.1 Anti Aliasing: UGC NET CSE | September 2013 | Part 3 | Question: 61**

Which of the following is/are fundamental method(s) of antialiasing?

- I. Increase of sample rate
II. Treating a pixel as a finite area rather than as a point
III. Decrease of sample rate
- A. I and II B. II and III C. I only D. II only

ugcnetcse-sep2013-paper3 computer-graphics anti-aliasing

Answer key **21.7 Bresenmans Algorithm (1)****21.7.1 Bresenmans Algorithm: UGC NET CSE | July 2016 | Part 3 | Question: 13**

Consider the Bresenman's circle generation algorithm for plotting a circle with centre (0,0) and radius 'r' unit in first quadrant. If the current point is (x_i, y_i) and decision parameter is p_i then what will be the next point (x_{i+1}, y_{i+1}) and updates decision parameter p_{i+1} for $p \geq 0$?

- A. $x_{i+1} = x_i + 1; y_{i+1} = y_i; p_{i+1} = p_i + 4x_i + 6$
B. $x_{i+1} = x_i + 1; y_{i+1} = y_i - 1; p_{i+1} = p_i + 4(x_i - y_i) + 10$
C. $x_{i+1} = x_i; y_{i+1} = y_i - 1; p_{i+1} = p_i + 4(x_i - y_i) + 6$
D. $x_{i+1} = x_i - 1; y_{i+1} = y_i; p_{i+1} = p_i + 4x_i + 10$

ugcnetcse-july2016-paper3 computer-graphics bresenmans-algorithm

Answer key 

21.8.1 Color Models: UGC NET CSE | September 2013 | Part 3 | Question: 62



The two color systems – the HSV and HLS are

- A. Hue, Saturation, Value and Hue, Lightness, Saturation
- B. High, Standard, Value and High Lightness, Saturation
- C. Highly, Saturated, Value and Highly, Lightened Saturation
- D. Highly, Standard, Value and Hue, Lightness, Saturation

ugcnetcse-sep2013-paper3 computer-graphics color-models

Answer key

21.9.1 Computer Graphics: Computer Science - UGC NET 2021 [Question ID = 2338]



What is the transformation matrix M that transforms a square in the xy plane defined by $(1, 1)$, $(-1, 1)$, $(-1, -1)$, $(1, -1)$ to a parallelogram whose corresponding vertices are $(2, 1)$, $(0, 1)$, $(-2, -1)$, and $(0, -1)$?

- | | |
|--|--|
| A. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | B. $\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |
| C. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ | D. $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ |

computer-graphics

Answer key

21.9.2 Computer Graphics: Computer Science - UGC NET 2021 [Question ID = 2339]



Suppose a Bezier curve $P(t)$ is defined by the following four control points in the xy - plane : $P_0 = (-2, 0)$; $P_1 = (-2, 4)$; $P_2 = (2, 4)$; and $P_3 = (2, 0)$. Then which of the following statements are correct ?

1. Bezier curve $P(t)$ has degree 3.
2. $P(1/2) = (0, 3)$.
3. Bezier curve $P(t)$ may extend outside the convex hull of its control points.

Choose the correct answer from the options given below :

- A. 1 and 2 only. B. 1 and 3 only. C. 2 and 3 only. D. 1, 2, and 3.

computer-graphics

21.9.3 Computer Graphics: Computer Science - UGC NET 2021 [Question ID = 2340]



Given below are two statements :

Statement I: The maximum number of sides that a triangle might have when clipped to a rectangular viewport is 6.

Statement II: In 3D graphics, the perspective transformation is nonlinear in z .

In light of the above statements, choose the correct answer from the options given below :

- A. Both Statement I and Statement II are true.
- B. Both Statement I and Statement II are false.
- C. Statement I is true but Statement II is false.
- D. Statement I is false but Statement II is true.

21.9.4 Computer Graphics: UGC NET CSE | December 2012 | Part 3 | Question: 12

Let $V_1 = 2I - J + k$ and $V_2 = I + J - K$, then the angle between V_1 and V_2 and a vector perpendicular to both V_1 and V_2 shall be

- A. 90° and $(-2I + J - 3K)$ B. 60° and $(2I + J + 3K)$
 C. 90° and $(2I + J - 3K)$ D. 90° and $(-2I - J + 3K)$

ugcnetcse-dec2012-paper3 computer-graphics

Answer key

21.9.5 Computer Graphics: UGC NET CSE | December 2012 | Part 3 | Question: 33

The Z-buffer algorithm is used for Hidden surface removal of objects. The maximum number of objects that can be handled by this algorithm shall

- A. depend on the application B. be arbitrary no. of objects
 C. depend on the memory availability D. depend on the processor

ugcnetcse-dec2012-paper3 computer-graphics

Answer key

21.9.6 Computer Graphics: UGC NET CSE | December 2012 | Part 3 | Question: 6

The aspect ratio of an image is defined as

- A. The ratio of width to its height measured in unit length
 B. The ratio of height to width measured in number of pixel
 C. The ratio of depth to width measured in unit length
 D. The ratio of width to depth measured in number of pixel

ugcnetcse-dec2012-paper3 computer-graphics

Answer key

21.9.7 Computer Graphics: UGC NET CSE | December 2013 | Part 3 | Question: 62

Which of the following statements is (are) true?

- I. Two successive translations are additive
 II. Two successive relations are additive
 III. Two successive scaling operations are multiplicative
- A. I and II B. I and III C. II and III D. All of the above

ugcnetcse-dec2013-paper3 computer-graphics non-gatecse

Answer key

21.9.8 Computer Graphics: UGC NET CSE | December 2013 | Part 3 | Question: 63

Given below are three basic rules:

- I. Squash and Stretch
 II. Slow-in and Slow-out
 III. To stage the action properly

These rules are applied in case of

- A. Rendering B. Morphing C. Animation D. All of the above

ugcnetcse-dec2013-paper3 computer-graphics non-gatecse

Answer key

21.9.9 Computer Graphics: UGC NET CSE | December 2013 | Part 3 | Question: 64



Which of the following points lies on the same side as the origin, with reference to the line $3x+7y=2$?

- A. (3, 0) B. (1, 0) C. (0.5, 0.5) D. (0.5, 0)

ugcnetcse-dec2013-paper3 computer-graphics non-gatecse

Answer key

21.9.10 Computer Graphics: UGC NET CSE | December 2013 | Part 3 | Question: 65



The transformation required for conversion of CMY colour model to RGB colour model is given as:

- A.
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$
- B.
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$
- C.
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} C \\ M \\ Y \end{bmatrix}$$
- D.
$$\begin{bmatrix} R \\ G \\ B \end{bmatrix} = \begin{bmatrix} C \\ M \\ Y \end{bmatrix} - \begin{bmatrix} 0.5 \\ 0.5 \\ 0.5 \end{bmatrix}$$

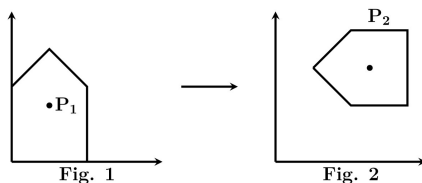
ugcnetcse-dec2013-paper3 computer-graphics non-gatecse

Answer key

21.9.11 Computer Graphics: UGC NET CSE | December 2013 | Part 3 | Question: 66



What steps shall be required to rotate an object from point P_1 (as shown in fig. 1) and its placement such that what was at P_1 is now reduced and is at P_2 (as shown in fig. 2)?



- I. Translate P_1 to origin
- II. Scale as required
- III. Rotate
- IV. Translate to the final position P_2

- A. I, II and III B. II, III and IV C. I, III and IV D. All of the above

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Answer key

21.9.12 Computer Graphics: UGC NET CSE | December 2014 | Part 3 | Question: 15



A technique used to approximate halftones without reducing spatial resolution is known as _____.

- A. Halftoning B. Dithering
C. Error diffusion D. None of the above

ugcnetcse-dec2014-paper3 computer-graphics non-gatecse

Answer key

21.9.13 Computer Graphics: UGC NET CSE | December 2015 | Part 3 | Question: 33

Which of the following graphic primitives are considered as the basic building blocks of computer graphics?

- i. Points
- ii. Lines
- iii. Polylines
- iv. Polygons

A. i only

B. i, ii, and iii

C. i and ii

D. i and iii

ugcnetcse-dec2015-paper3 computer-graphics

Answer key

21.9.14 Computer Graphics: UGC NET CSE | December 2015 | Part 3 | Question: 50

Which of the following is not a lossy compression technique?

A. JPEG

B. MPEG

C. FFT

D. Arithmetic coding

ugcnetcse-dec2015-paper3 computer-graphics

Answer key

21.9.15 Computer Graphics: UGC NET CSE | December 2015 | Part 3 | Question: 67

Which of the following is/are the principle components of a memory-tube display?

- i. Flooding gun
- ii. Collector
- iii. Phosphorus grains
- iv. Ground

A. i and ii

B. iii only

C. iv only

D. All of the above

ugcnetcse-dec2015-paper3 computer-graphics

Answer key

21.9.16 Computer Graphics: UGC NET CSE | December 2015 | Part 3 | Question: 69

Which raster locations would be chosen by Bresenham's algorithm when scan converting a line from (1,1) to (8,5)?

A.

x	y
1	1
2	2
3	3
4	3
5	4
6	4
7	5
8	6

B.

x	y
1	1
2	2
3	2
4	3
5	4
6	5
7	6
8	7

C.

x	y
1	1
2	2
3	2
4	3
5	3
6	4
7	4
8	5

D.

x	y
1	1
2	2
3	2
4	3
5	5
6	4
7	5
8	5

21.9.17 Computer Graphics: UGC NET CSE | December 2015 | Part 3 | Question: 70



Consider a unit square centered at origin. The coordinates at the square are translated by a factor $\left(\frac{1}{2}, 1\right)$ and rotated by an angle of 90° . What shall be the coordinates of the new square?

A.

$$\left(\frac{-1}{2}, 0\right), \left(\frac{-1}{2}, 1\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$$

B.

$$\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 1\right), \left(\frac{3}{2}, 1\right), \left(\frac{3}{2}, 0\right)$$

C.

$$\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 0\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$$

D.

$$\left(\frac{-1}{2}, 0\right), \left(\frac{1}{2}, 1\right), \left(\frac{-3}{2}, 1\right), \left(\frac{-3}{2}, 0\right)$$

Answer key 

21.9.18 Computer Graphics: UGC NET CSE | January 2017 | Part 3 | Question: 16



Below are the few steps given for scan-converting a circle using Bresenham's Algorithm. Which of the given steps is not correct?

- A. Compute $d = 3 - 2r$ (where r is radius)
C. If $d < 0$, then $d = 4x + 6$ and $x = x + 1$

- B. Stop if $x > y$
D. If $d \geq 0$, then $d = 4 * (x - y) + 10$, $x = x + 1$ and $y = y + 1$

ugcnetcse-jan2017-paper3 non-gatecse computer-graphics

Answer key 

21.9.19 Computer Graphics: UGC NET CSE | January 2017 | Part 3 | Question: 17



Which of the following is/are side effects of scan conversion?

- i. Aliasing
ii. Unequal intensity of diagonal lines
iii. Overstriking in photographic applications
iv. Local or Global aliasing

- A. i and ii
C. i,iii and iv

- B. i,ii and iii
D. i,ii,iii and iv

ugcnetcse-jan2017-paper3 non-gatecse computer-graphics

Answer key 

21.9.20 Computer Graphics: UGC NET CSE | January 2017 | Part 3 | Question: 18



Consider a line AB with $A = (0, 0)$ and $B = (8, 4)$. Apply a simple DDA algorithm and compute the first four plots on this line.

- A. $[(0, 0), (1, 1), (2, 1), (3, 2)]$
C. $[(0, 0), (1, 1), (2, 2.5), (3, 3)]$

- B. $[(0, 0), (1, 1.5), (2, 2), (3, 3)]$
D. $[(0, 0), (1, 2), (2, 2), (3, 2)]$

ugcnetcse-jan2017-paper3 non-gatecse computer-graphics

Answer key 

21.9.21 Computer Graphics: UGC NET CSE | July 2016 | Part 3 | Question: 15



Let R be the rectangular window against which the lines are to be clipped using 2D Sutherland-Cohen line clipping algorithm. The rectangular window has lower rectangular corner at $(-5, 1)$ and upper right-hand corner at $(3, 7)$. Consider the following three lines for clipping with the given end-point coordinates?

Line AB: $A(-6, 2)$ and $B(-1, 8)$

Line CD: $C(-1, 5)$ and $D(4, 8)$

Line EF: $E(-2, 3)$ and $F(1, 2)$

Which of the following lines is/are candidate for clipping?

- A. AB B. DC C. EF D. AB and CD

ugcnetcse-july2016-paper3 computer-graphics

Answer key 

21.9.22 Computer Graphics: UGC NET CSE | July 2016 | Part 3 | Question: 17



Let us consider that the original point is (x, y) and new transformed point is (x', y') . Further Sh_x and Sh_y are shearing factors in x and y directions. If we perform the y direction shear relative to $x = x_{ref}$ then the transformed point is given by

- A. $x' = x + Sh_x \cdot (y - y_{ref}); y' = y$
C. $x' = x; y' = Sh_y \cdot (x - x_{ref}) + y$

- B. $x' = x; y' = y \cdot Sh_x$
D. $x' = Sh_y \cdot y; y' = y \cdot (x - x_{ref})$

ugcnetcse-july2016-paper3 computer-graphics

Answer key 

21.9.23 Computer Graphics: UGC NET CSE | July 2016 | Part 3 | Question: 18



Which of the following statements is/are correct with reference to curve generation?

- I. Hermite curves are generated using the concepts of interpolation
 - II. Bezier curves are generated using the concepts of approximation
 - III. The Bezier curve lies entirely within the convex hull of its control points
 - IV. The degree of Bezier curve does not depend on the number of control points
- A. I, II and IV only B. II and III only C. I and II only D. I, II and III only

ugcnetcse-july2016-paper3 computer-graphics

Answer key 

21.9.24 Computer Graphics: UGC NET CSE | July 2018 | Part 2 | Question: 8



A graphic display system has a frame buffer that is 640 pixels high , 480 pixels high and 1 bit of color depth. If the access time for each pixel on the average is 200 nanoseconds, then the refresh rate of this frame buffer is approximately:

- A. 16 frames per second B. 19 frames per second
C. 21 frames per second D. 23 frames per second

ugcnetcse-july2018-paper2 computer-graphics

Answer key 

21.9.25 Computer Graphics: UGC NET CSE | June 2012 | Part 3 | Question: 14



If the pixels of an image are shuffled then the parameter that may change is

- A. Histogram B. Mean C. Entropy D. Covariance

ugcnetcse-june2012-paper3 computer-graphics

Answer key 

21.9.26 Computer Graphics: UGC NET CSE | June 2012 | Part 3 | Question: 63



If a and b are the end points of a line, then which of the following is true?

- A. If both end points are left, right, above or below the window, the line is invisible.
B. If both end points are left, right, above or below the window, the line is completely visible.
C. If both end points are left, right, above or below the window, the line is trivially visible
D. If both end points are left, right, above or below the window, the line is trivially invisible

computer-graphics non-gatecse ugcnetcse-june2012-paper3

Answer key 

21.9.27 Computer Graphics: UGC NET CSE | June 2014 | Part 3 | Question: 01



Beam-penetration and shadow-mask are the two basic techniques for producing color displays with a CRT. Which of the following is not true ?

- I. The beam-penetration is used with random scan monitors.
II. Shadow-mask is used in rasterscan systems.
III. Beam-penetration method is better than shadow-mask method.
IV. Shadow-mask method is better than beam-penetration method.

- A. I and II B. II and III C. III only D. IV only

ugcnetjune2014iii non-gatecse computer-graphics

Answer key 

21.9.28 Computer Graphics: UGC NET CSE | June 2014 | Part 3 | Question: 02



Line caps are used for adjusting the shape of the line ends to give them a better appearance. Various kinds of line caps used are

- A. Butt cap and sharp cap
- B. Butt cap and round cap
- C. Butt cap, sharp cap and round cap
- D. Butt cap, round cap and projecting square cap

ugcnetjune2014iii non-gatecse computer-graphics

Answer key 

21.9.29 Computer Graphics: UGC NET CSE | June 2014 | Part 3 | Question: 03



Given below are certain output primitives and their associated attributes. Match each primitive with its corresponding attributes :

List – I

List – II

- | | |
|--------------|----------------------------|
| a. Line | i. Type, Size, Color |
| b. Fill Area | ii. Color, Size, Font |
| c. Text | iii. Style, Color, Pattern |
| d. Marker | iv. Type, Width, Color |

Codes :

- A. a-i, b-ii, c-iii, d-iv
- B. a-ii, b-i, c-iii, d-iv
- C. a-iv, b-iii, c-ii, d-i
- D. a-iii, b-i, c-iv, d-ii

ugcnetjune2014iii non-gatecse computer-graphics

Answer key 

21.9.30 Computer Graphics: UGC NET CSE | June 2014 | Part 3 | Question: 04



Consider a window bounded by the lines : $x = 0$; $y = 0$; $x = 5$ and $y = 3$. The line segment joining $(-1, 0)$ and $(4, 5)$, if clipped against this window will connect the points

- A. $(0, 1)$ and $(2, 3)$
- B. $(0, 1)$ and $(3, 3)$
- C. $(0, 1)$ and $(4, 3)$
- D. $(0, 1)$ and $(3, 2)$

ugcnetjune2014iii non-gatecse geometry computer-graphics

Answer key 

21.9.31 Computer Graphics: UGC NET CSE | June 2014 | Part 3 | Question: 05



Which of the following color models are defined with three primary colors ?

- A. RGB and HSV color models
- B. CMY and HSV color models
- C. HSV and HLS color models
- D. RGB and CMY color models

ugcnetjune2014iii non-gatecse computer-graphics

Answer key 

21.9.32 Computer Graphics: UGC NET CSE | Junet 2015 | Part 3 | Question: 16



Which of the following is not a basic primitive of the Graphics Kernel System (GKS)?

- A. POLYLINE
- B. POLYDRAW
- C. FILL AREA
- D. POLYMARKER

ugcnetcse-june2015-paper3 computer-graphics

Answer key 

21.9.33 Computer Graphics: UGC NET CSE | Junet 2015 | Part 3 | Question: 17



Which of the following statements is/are incorrect?

- Mapping the co-ordinates of the points and lines that form the picture into the appropriate co-ordinates on the device or workstation is known as viewing transformation
- The right-handed cartesian co-ordinates system in whose co-ordinates we describe the picture is known as world co-ordinate system
- The co-ordinate system that corresponds to the device or workstation where the image is to be displayed is known as physical device co-ordinate system
- Left-handed co-ordinate system in which the display area of the virtual display device corresponds to the unit $(|x|)$ square whose lower left-hand corner is at the origin of the co-ordinate system, is known as normalized device co-ordinate system

A. i only

B. i and ii

C. iii only

D. iv only

ugcnetcse-june2015-paper3 computer-graphics

Answer key 

21.9.34 Computer Graphics: UGC NET CSE | Junet 2015 | Part 3 | Question: 18



Match the following :

List-I

(a) Flood Gun

(b) Collector

(c) Ground

(d) Phosphorous Grains

(e) Writing Gun System

List-II

(i) An electron gun designed to flood the entire screen with electrons.

(ii) Partly energized by flooding gun, stores the charge generated by the writing gun.

(iii) Used to discharge the collector.

(iv) Used in memory - tube display and similar to those used in standard CRT.

(v) Used in memory - tube display and basically the same as the electron gun used in a conventional CRT.

Codes :

A. (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv), (e)-(v)

B. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv), (e)-(v)

C. (a)-(iii), (b)-(i), (c)-(ii), (d)-(v), (e)-(iv)

D. (a)-(iv), (b)-(v), (c)-(i), (d)-(ii), (e)-(iii)

ugcnetcse-june2015-paper3 computer-graphics

Answer key 

21.9.35 Computer Graphics: UGC NET CSE | November 2017 | Part 3 | Question: 13



Which of the following is not a component of Memory tube display?

A. Flooding gun

B. Collector

C. Ground

D. Liquid Crystal

ugcnetcse-nov2017-paper3 computer-graphics

Answer key 

21.9.36 Computer Graphics: UGC NET CSE | October 2020 | Part 2 | Question: 46



In the context of 3D Computer graphics, which of the following statements is/are correct?

- Under perspective projection, each set of parallel lines in the object do not stay parallel in the image (except those that are parallel to the viewplane to start with).
- Applying a perspective transformation in the graphics pipeline to a vertex involves dividing by its $'z'$ coordinate
- Perspective transformation is a linear transformation

Choose the correct answer from the options given below:

- A. (i) and (ii) only
C. (ii) and (iii) only

- B. (i) and (iii) only
D. (i), (ii) and (iii)

ugcnetcse-oct2020-paper2 non-gatecse computer-graphics

Answer key

21.9.37 Computer Graphics: UGC NET CSE | October 2020 | Part 2 | Question: 47



Concerning phong shading and Gouraud shading in a 3D scene, which of the following statements are true?

- Gouraud shading requires more computation than phong shading
- Gouraud shading linearly interpolates the color of an interior pixel from the color at the vertices
- Phong shading interpolates over the normal vectors specified at the vertices

Choose the correct answer from the options given below:

- A. (i) and (ii) only
C. (ii) and (iii) only

- B. (i) and (iii) only
D. (i), (ii) and (iii)

ugcnetcse-oct2020-paper2 non-gatecse computer-graphics

21.9.38 Computer Graphics: UGC NET CSE | October 2020 | Part 2 | Question: 66



Match List I with List II. List I gives 3×3 matrices representing 2D transformations and List II shows the corresponding transformation diagrams.

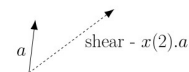
List I (Matrix)	List II (Transformation Diagram)
(A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(I)
(B) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$	(II)
(C) $\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(III)
(D) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	(IV)

List I (Matrix)

List II (Transformation Diagram)

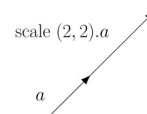
(A) $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(I)



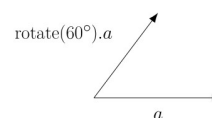
(B) $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$

(II)



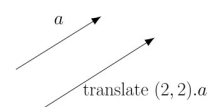
(C) $\begin{bmatrix} 0.5 & -0.866 & 0 \\ 0.866 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(III)



(D) $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(IV)



Choose the correct answer from the options given below:

- A. $A - IV, B - II, C - III, D - I$
C. $A - III, B - II, C - IV, D - I$

- B. $A - IV, B - III, C - II, D - I$
D. $A - II, B - IV, C - III, D - I$

ugcnetcse-oct2020-paper2 non-gatecse computer-graphics

Answer key 

21.9.39 Computer Graphics: UGC NET CSE | October 2020 | Part 2 | Question: 76



Given below are different properties of 3D projections from A-D. Identify the correct order on the basis of property true of (i) a perspective projection only, (ii) an orthographic projection only, (iii) both orthographic and projective transformations and (iv) neither orthographic nor projective transformation, respectively.

- i. Straight lines are mapped to straight lines
- ii. Distances and angles are (in general) preserved
- iii. Far away objects appear the same size as closer ones
- iv. Requires homogeneous coordinates in order for it to be encoded into a linear transformation

Choose the correct answer from the options given below:

A. iv,iii,ii,i

B. ii,iii,iv,i

C. iv,iii,i,ii

D. iii,iv,ii,i

ugcnetcse-oct2020-paper2 non-gatecse computer-graphics

Answer key 

21.9.40 Computer Graphics: UGC NET CSE | October 2020 | Part 2 | Question: 82



Given below are two statements:

Statement *I*: Bezier curves are curves that interpolate all of their control points

Statement *II*: A cubic bezier curve has four control points

In the light of the above statements, choose the correct answer from the options given below

- A. Both Statement *I* and Statement *II* are true
- B. Both Statement *I* and Statement *II* are false
- C. Statement *I* is correct but Statement *II* is false
- D. Statement *I* is incorrect but Statement *II* is true

ugcnetcse-oct2020-paper2 non-gatecse computer-graphics

Answer key 

21.9.41 Computer Graphics: UGC NET CSE | September 2013 | Part 3 | Question: 59



Which of the following graphics devices are known as active graphics devices?

- I. Alphanumeric devices
- II. Thumb wheels
- III. Digitizers
- IV. Joysticks

A. I and II

B. III and IV

C. I, II and III

D. I, II, III and IV

ugcnetcse-sep2013-paper3 computer-graphics

Answer key 

21.10

Crt (2)

21.10.1 Crt: UGC NET CSE | December 2015 | Part 3 | Question: 71



Which of the following is/are the components of a CRT?

- i. Electron Gun
- ii. Control Electrode
- iii. Focusing Electrode
- iv. Phosphor Coated Screen

- A. i and iv
- C. i, ii, iii and iv

- B. i, ii and iv
- D. i, iii and iv

ugcnetcse-dec2015-paper3 computer-graphics crt

Answer key 

21.10.2 Crt: UGC NET CSE | November 2017 | Part 3 | Question: 15



With respect to CRT, the horizontal retrace is defined as

- A. The path an electron beam takes when returning to the left side of the CRT
- B. The path an electron beam takes when returning to the right side of the CRT
- C. The technique of turning, the electron beam off while retracing
- D. The technique of turning, the electron beam on/ off while retracing

ugcnetcse-nov2017-paper3 computer-graphics crt

Answer key 

21.11 Digital Image Processing (5)

21.11.1 Digital Image Processing: UGC NET CSE | June 2013 | Part 3 | Question: 19



An image is 1024*800 pixels with 3 bytes/pixel. Assume the image is uncompressed. How long does it take to transmit it over a 10 Mbps Ethernet?

- A. 196.6 seconds
- B. 19.66 seconds
- C. 1.966 seconds
- D. 0.1966 seconds

ugcnetcse-june2013-paper3 computer-graphics digital-image-processing

Answer key 

21.11.2 Digital Image Processing: UGC NET CSE | June 2013 | Part 3 | Question: 31



Images tend to be very large collection of data. The size of memory required for a 1024 by 1024 image in which the color of each pixel is represented by a n-bit number, (in an 8-bit machines) is

- A. $n \times 8$ MB
- B. $n/8$ MB
- C. $(1024 \times 1024)/8$ MB
- D. 1024 MB

ugcnetcse-june2013-paper3 computer-graphics digital-image-processing

Answer key 

21.11.3 Digital Image Processing: UGC NET CSE | June 2013 | Part 3 | Question: 56



The number of distinct binary images which can be generated from a given binary image of right $M \times N$ are

- A. $M+N$
- B. $M \times N$
- C. 2^{M+N}
- D. 2^{MN}

ugcnetcse-june2013-paper3 computer-graphics digital-image-processing

Answer key 

21.11.4 Digital Image Processing: UGC NET CSE | June 2013 | Part 3 | Question: 57



If $f(x, y)$ is a digital image, then x, y and amplitude values of f are

- A. Finite
- B. Infinite
- C. Neither finite nor infinite
- D. None of the above

ugcnetcse-june2013-paper3 computer-graphics digital-image-processing

Answer key 

21.11.5 Digital Image Processing: UGC NET CSE | November 2017 | Part 3 | Question: 18



The three aspects of Quantization, programmers generally concerned with are

- A. Coding error, Sampling rate and Amplification
- B. Sampling rate, Coding error and Conditioning
- C. Sampling rate, Aperture time and Coding error
- D. Aperture time, Coding error and Strobing

ugcnetcse-nov2017-paper3 computer-graphics digital-image-processing

Answer key

21.12 Display Devices (1)

21.12.1 Display Devices: UGC NET CSE | June 2013 | Part 3 | Question: 28



The refresh rate above which a picture stops flickering and fuses into a steady image is called ____

- A. Crucial fusion frequency
- B. Current frequency fusion
- C. Critical fusion frequency
- D. Critically diffused frequency

ugcnetcse-june2013-paper3 computer-graphics display-devices

Answer key

21.13 Frame Buffer (1)

21.13.1 Frame Buffer: UGC NET CSE | August 2016 | Part 3 | Question: 14



Consider a N -bit plane frame buffer with W -bit wide lookup table with $W > N$. How many intensity levels are available at a time ?

- A. 2^N
- B. 2^W
- C. 2^{N+W}
- D. 2^{N-1}

ugcnetcse-aug2016-paper3 computer-graphics frame-buffer

Answer key

21.14 Halftoning (1)

21.14.1 Halftoning: UGC NET CSE | June 2012 | Part 3 | Question: 75



Halftoning is defined as

- A. A technique to obtain increased visual resolution using multiple intensity levels
- B. A technique for using minimum number of intensity levels to obtain increased visual resolution
- C. A technique to obtain increased visual resolution using maximum number of intensity levels
- D. A technique for using appropriate number intensity levels to obtain increased visual resolution

ugcnetcse-june2012-paper3 computer-graphics halftoning

Answer key

21.15 Image Compression (1)

21.15.1 Image Compression: UGC NET CSE | September 2013 | Part 3 | Question: 55



MPEG involves both spatial compression and temporal compression. The spatial compression is similar to JPEG and temporal compression removes ____ frames.

- A. Temporal
- B. Voice
- C. Spatial
- D. Redundant

ugcnetcse-sep2013-paper3 computer-graphics digital-image-processing image-compression

Answer key

21.16

Image Entropy (1)

21.16.1 Image Entropy: UGC NET CSE | December 2012 | Part 3 | Question: 71



You are given four images to represent as

$$I_1 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, I_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, I_3 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, I_4 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

The value of entropy is maximum for image

- A. I_1 B. I_2 C. I_3 D. I_4

ugcnetcse-dec2012-paper3 computer-graphics image-entropy

Answer key

21.17

Image Rotation (2)

21.17.1 Image Rotation: UGC NET CSE | July 2016 | Part 3 | Question: 14



A point P(5,1) is rotated by 90° about a pivot point (2,2). What is the coordinate of new transformed point P'?

- A. (3,5) B. (5,3) C. (2,4) D. (1,5)

ugcnetcse-july2016-paper3 computer-graphics image-rotation

Answer key

21.17.2 Image Rotation: UGC NET CSE | Junet 2015 | Part 3 | Question: 14



Consider a triangle A(0,0), B(1, 1) and C(5, 2). The triangle has to be rotated by an angle of 45° about the point P(-1, -1). What shall be the coordinates of new triangle?

- A. $A' = (1, \sqrt{2} - 1), B'(-1, 2\sqrt{2} - 1)$ and $C' = (3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1)$
 B. $A' = (1, \sqrt{2} - 1), B'(2\sqrt{2} - 1, -1)$ and $C' = (3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1)$
 C. $A' = (-1, \sqrt{2} - 1), B'(-1, 2\sqrt{2} - 1)$ and $C' = (3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1)$
 D. $A' = (\sqrt{2} - 1, -1), B'(-1, 2\sqrt{2} - 1)$ and $C' = (3\sqrt{2} - 1, \frac{9}{2}\sqrt{2} - 1)$

ugcnetcse-june2015-paper3 computer-graphics image-rotation

Answer key

21.18

Line Clipping (1)

21.18.1 Line Clipping: UGC NET CSE | December 2014 | Part 3 | Question: 17



In Cyrus-Beck algorithm for line clipping the value of t parameter is computed by the relation:

(Here P_1 and P_2 are the two end points of the line, f is a point on the boundary, n_1 is inner normal).

- A. $\frac{(P_1 - f) \cdot n_i}{(P_2 - P_1) \cdot n_i}$ B. $\frac{(f - P_1) \cdot n_i}{(P_2 - P_1) \cdot n_i}$
 C. $\frac{(P_2 - f) \cdot n_i}{(P_1 - P_2) \cdot n_i}$ D. $\frac{(f - P_2) \cdot n_i}{(P_1 - P_2) \cdot n_i}$

ugcnetcse-dec2014-paper3 non-gatecse computer-graphics line-clipping

Answer key

21.19

Line Drawing Algorithm (3)

21.19.1 Line Drawing Algorithm: UGC NET CSE | August 2016 | Part 3 | Question: 15



Consider the Bresenham's line generation algorithm for a line with gradient greater than one, current point (x_i, y_i) and decision parameter, d_i . The next point to be plotted (x_{i+1}, y_{i+1}) and updated decision parameter, d_{i+1} , for $d_i < 0$ are given as _____.

- A. $x_{i+1} = x_i y_{i+1} = y_i d_{i+1} = d_i + 2dy$
 B. $x_{i+1} = x_i y_{i+1} = y_{i+1} d_{i+1} = d_i + 2dx$
 C. $x_{i+1} = x_i y_{i+1} = y_{i+1} d_{i+1} = d_i + 2(dx - dy)$
 D. $x_{i+1} = x_{i+1} y_{i+1} = y_{i+1} d_{i+1} = d_i + 2(dy - dx)$

ugcnetcse-aug2016-paper3 computer-graphics line-drawing-algorithm

Answer key 

21.19.2 Line Drawing Algorithm: UGC NET CSE | August 2016 | Part 3 | Question: 18



The Liang-Barsky line clipping algorithm uses the parametric equation of a line from (x_1, y_1) to (x_2, y_2) along with its infinite extension which is given as :

$$x = x_1 + \Delta x \cdot u$$

$$y = y_1 + \Delta y \cdot u$$

Where $\Delta x = x_2 - x_1$, $\Delta y = y_2 - y_1$, and u is the parameter with $0 \leq u \leq 1$. A line AB with end points $A(-1, 7)$ and $B(11, 1)$ is to be clipped against a rectangular window with $x_{min} = 1, x_{max} = 9, y_{min} = 2$, and $y_{max} = 8$. The lower and upper bound values of the parameter u for the clipped line using Liang-Barsky algorithm is given as :

- A. $(0, \frac{2}{3})$ B. $(\frac{1}{6}, \frac{5}{6})$ C. $(0, \frac{1}{3})$ D. $(0, 1)$

ugcnetcse-aug2016-paper3 computer-graphics line-drawing-algorithm

Answer key 

21.19.3 Line Drawing Algorithm: UGC NET CSE | June 2013 | Part 3 | Question: 27



Bresenham line drawing algorithm is attractive because it uses

- A. Real arithmetic only B. Integer arithmetic only
 C. Floating point arithmetic D. Real and integer arithmetic

ugcnetcse-june2013-paper3 computer-graphics line-drawing-algorithm

Answer key 

21.20 Parametric Line Equation (1)

21.20.1 Parametric Line Equation: UGC NET CSE | September 2013 | Part 3 | Question: 63



The parametric representation of the line segment between the position vectors $P_1(2, 3)$ and $P_2(5, 4)$ is given as

- A. $x(t) = 2 + 7t, y(t) = 0 \leq t \leq \infty \quad 3 + 7t$
 B. $x(t) = 2 + 10t, y(t) = 0 \leq t \leq 1 \quad 3 + 7t$
 C. $x(t) = 2 + 3t, y(t) = 0 \leq t \leq 1 \quad y(t) = 3 + t$
 D. $t(x, y) = 14t \quad 0 \leq t \leq 1$

ugcnetcse-sep2013-paper3 computer-graphics parametric-line-equation

Answer key 

21.21 Phong Color Model (1)

21.21.1 Phong Color Model: UGC NET CSE | June 2019 | Part 2 | Question: 30



Using the phong reflectance model, the strength of the specular highlight is determined by the angle between

- A. the view vector and the normal vector B. the light vector and the normal vector
 C. the light vector and the reflected vector D. the reflected vector and the view vector

ugcnetcse-june2019-paper2 phong-color-model

Answer key 

21.22 Projection (8)

21.22.1 Projection: UGC NET CSE | August 2016 | Part 3 | Question: 16

A point $P(2,5)$ is rotated about a pivot point $(1,2)$ by 60° . What is the new transformed point P' ?

- A. $(1,4)$ B. $(-1,4)$ C. $(1,-4)$ D. $(-4,1)$

ugcnetcse-aug2016-paper3 computer-graphics projection

Answer key

21.22.2 Projection: UGC NET CSE | August 2016 | Part 3 | Question: 17

In perspective projection (from 3D to 2D), objects behind the centre of projection are projected upside down and backward onto the view-plane. This is known as ____.

- A. Topological distortion B. Vanishing point
C. View confusion D. Perspective foreshortening

ugcnetcse-aug2016-paper3 computer-graphics projection

Answer key

21.22.3 Projection: UGC NET CSE | December 2014 | Part 3 | Question: 18

Match the following :

List – I**List – II**

- | | |
|-------------------------------|---|
| a. Cavalier
Projection | i. The direction of projection is chosen so that there is no foreshortening of lines perpendicular to the xy plane. |
| b. Cabinet
Projection | ii. The direction of projection is chosen so that lines perpendicular to the xy planes are foreshortened by half their lengths. |
| c. Isometric
Projection | iii. The direction of projection makes equal angles with all of the principal axis. |
| d. Orthographic
Projection | iv. Projections are characterized by the fact that the direction of projection is perpendicular to the view plane. |

Codes :

- A. a-i, b-iii, c-iv, d-ii B. a-ii, b-iii, c-i, d-iv
C. a-iv, b-ii, c-iii, d-i D. a-i, b-ii, c-iii, d-iv

ugcnetcse-dec2014-paper3 computer-graphics projection non-gatecse

Answer key

21.22.4 Projection: UGC NET CSE | July 2016 | Part 3 | Question: 16

In perspective projection, if a line segment joining a point which lies in front of the viewer to a point in back of the viewer is projected to a broken line of infinite extent. This is known as ____

- A. View confusion B. Vanishing point
C. Topological distortion D. Perspective foreshortening

ugcnetcse-july2016-paper3 computer-graphics projection

Answer key

21.22.5 Projection: UGC NET CSE | June 2012 | Part 3 | Question: 59

The perspective projection matrix, on the view plan $z=d$ where the centre of projection is the origin $(0, 0, 0)$ shall be

A.
$$\begin{bmatrix} 0 & 0 & 0 & d \\ 0 & 0 & d & 0 \\ 0 & d & 0 & 0 \\ d & 0 & 0 & 1 \end{bmatrix}$$

B.
$$\begin{bmatrix} d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

C.
$$\begin{bmatrix} 0 & 0 & 0 & d \\ 0 & 0 & 0 & d \\ 0 & 0 & d & 0 \\ 0 & d & 0 & 0 \end{bmatrix}$$

D.
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ d & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & d & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

ugcnetcse-june2012-paper3 computer-graphics projection

Answer key 

21.22.6 Projection: UGC NET CSE | June 2013 | Part 3 | Question: 29



In homogenous coordinate system (x, y, z) the points $z=0$ are called

- A. Cartesian points
B. Parallel points
C. Origin point
D. Point of infinity

ugcnetcse-june2013-paper3 computer-graphics projection

Answer key 

21.22.7 Projection: UGC NET CSE | November 2017 | Part 3 | Question: 14



Which of the following is **not** true in case of Oblique Projections?

- A. Parallel projection rays are not perpendicular to the viewing plane
B. Parallel lines in space appear parallel on the final projected image
C. Used exclusively for pictorial purposes rather than formal working drawings
D. Projectors are always perpendicular to the plane of projection

ugcnetcse-nov2017-paper3 computer-graphics projection

Answer key 

21.22.8 Projection: UGC NET CSE | September 2013 | Part 3 | Question: 60



A diametric projection is said to be trimetric projection when

- I. two of the three foreshortening factors are equal and third is arbitrary
II. all of the three foreshortening factors are equal
III. all of the three foreshortening factors are arbitrary

Which of the above is true?

- A. I and II
B. II and III
C. I only
D. III only

ugcnetcse-sep2013-paper3 computer-graphics projection

Answer key 

21.23.1 Raster Graphics: UGC NET CSE | August 2016 | Part 3 | Question: 13

Consider a raster grid having XY -axes in positive X -direction and positive upward Y -direction with $X_{max} = 10$, $X_{min} = -5$, $Y_{max} = 11$, and $Y_{min} = 6$. What is the address of memory pixel with location $(5, 4)$ in raster grid assuming base address 1 (one) ?

- A. 150 B. 151 C. 160 D. 161

ugcnetcse-aug2016-paper3 computer-graphics raster-graphics

Answer key

21.23.2 Raster Graphics: UGC NET CSE | June 2019 | Part 2 | Question: 27

Consider a raster system with resolution 640 by 480. What size is frame buffer (in bytes) for this system to store 12 bits per pixel?

- A. 450 kilobytes B. 500 kilobytes
C. 350 kilobytes D. 400 kilobytes

ugcnetcse-june2019-paper2 raster-graphics

Answer key

21.24 Resolution (1)**21.24.1 Resolution: UGC NET CSE | June 2013 | Part 3 | Question: 30**

If 40 black lines interleaved with 40 white lines can be distinguished across one inch, the resolution is

- A. 40 line-pairs per inch B. 80 line-pairs per inch
C. 1600 lines per inch D. 40 lines per inch

ugcnetcse-june2013-paper3 computer-graphics resolution

Answer key

21.25 Transformation (2)**21.25.1 Transformation: UGC NET CSE | December 2012 | Part 3 | Question: 16**

The Mandelbrot set used for the construction of beautiful images is based on the following transformation.
 $x_{n+1} = x_n^2 + z$ Here,

- A. Both x and z are real numbers B. Both x and z are complex numbers
C. x is real and z is complex D. x is complex and z is real

ugcnetcse-dec2012-paper3 computer-graphics transformation

Answer key

21.25.2 Transformation: UGC NET CSE | Junet 2015 | Part 3 | Question: 13

Give the number of principal vanishing point(s) along with their direction for the standard prospective transformation:

- A. Only one in the direction K B. Two in the directions i and j
C. Three in the directions I, J and K D. Only two in the directions J and K

ugcnetcse-june2015-paper3 computer-graphics transformation

Answer key

Answer Keys

21.1.1	Q-Q	21.2.1	B	21.3.1	Q-Q	21.3.2	A	21.4.1	C
21.5.1	D	21.6.1	A	21.7.1	B	21.8.1	A	21.9.1	Q-Q
21.9.2	Q-Q	21.9.3	Q-Q	21.9.4	X	21.9.5	B	21.9.6	A

21.9.7	D
21.9.12	B
21.9.17	D
21.9.22	C
21.9.27	C
21.9.32	B
21.9.37	C
21.10.1	C
21.11.4	A
21.15.1	D
21.19.1	Q-Q
21.22.1	Q-Q
21.22.6	D
21.24.1	A

21.9.8	C
21.9.13	D
21.9.18	D
21.9.23	D
21.9.28	D
21.9.33	D
21.9.38	D
21.10.2	A
21.11.5	C
21.16.1	C
21.19.2	Q-Q
21.22.2	Q-Q
21.22.7	D
21.25.1	B

21.9.9	D
21.9.14	X
21.9.19	D
21.9.24	Q-Q
21.9.29	C
21.9.34	A
21.9.39	C
21.11.1	C
21.12.1	C
21.17.1	A
21.19.3	B
21.22.3	D
21.22.8	C
21.25.2	A

21.9.10	C
21.9.15	B
21.9.20	A
21.9.25	D
21.9.30	A
21.9.35	D
21.9.40	D
21.11.2	B
21.13.1	Q-Q
21.17.2	C
21.20.1	C
21.22.4	C
21.23.1	Q-Q

21.9.11	X
21.9.16	C
21.9.21	D
21.9.26	D
21.9.31	D
21.9.36	A
21.9.41	D
21.11.3	D
21.14.1	B
21.18.1	B
21.21.1	D
21.22.5	B
21.23.2	A